Simple Guide: Distributed Food Ordering

Kafka (KRaft) + Spark Batch · Big Data 2026 (UE24CS343AB2) · Team of 4: Member 1 = VM1, Member 2 = VM2, Member 3 = VM3, Member 4 = VM4

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1. The whole assignment in one minute

A food delivery app has a file of **62 orders** (`order.jsonl`). Some of them are junk: broken records, cancelled orders, bad timestamps, duplicates. We have to:

- **Send** the raw orders through Kafka (VM1 -> VM4).
- **Clean** them and add delivery numbers with Spark (VM4). 62 raw -> **44 good orders**.
- **Send** the cleaned orders back through Kafka (VM4 -> VM1).
- **Summarise** them with Spark (VM1) into two reports: one per restaurant, one per city.
- **Prove** that Kafka keeps working when one broker (VM3) is switched off.

Analogy. Kafka is a **post office** that stores letters (messages) in labelled mailboxes (*topics*). It keeps two copies of every letter, one in each branch (VM2 and VM3), so if one branch closes the letters are still there. Spark is the **calculator** each side uses on its own desk after collecting the mail.

The 4 machines

Member / VM	Role	What it does in plain words
Member 1 / VM1	Producer 1, Consumer 2, Spark Job 2	Posts the raw orders into Kafka. Later collects the cleaned orders and makes the two final CSV reports.
Member 2 / VM2	Kafka broker 1 + controller	A Kafka server, and also the **boss** of the cluster: it knows which servers are alive and who is in charge of each piece of data.
Member 3 / VM3	Kafka broker 2	The second Kafka server. Holds a backup copy of everything. This is the one we switch off in the failure test.
Member 4 / VM4	Consumer 1, Spark Job 1, Producer 2	Collects the raw orders, cleans them and adds metrics with Spark, then posts the cleaned orders back into Kafka.

The flow, step by step

```
STAGE 1                                      STAGE 2
VM1  producer1.py                            VM4  producer2.py
      |  topic: orders  (62 msgs)                   |  topic: order_metrics (44 msgs)
      v                                             v
VM4  consumer1.py -> orders_landed.jsonl     VM1  consumer2.py -> order_metrics_landed.jsonl
      v                                             v
VM4  spark_job1.py -> order_metrics.jsonl    VM1  spark_job2.py -> restaurant_summary.csv
      (clean + enrich, 62 -> 44)                                  -> city_summary.csv

Both topics live on the Kafka cluster = VM2 + VM3 (connected over ZeroTier)
Each topic: 2 partitions, replication factor 2
```

Rules from the spec (the things examiners check)

- Spark never reads Kafka directly. A consumer first saves the data to a local `.jsonl` file; Spark reads that file.
- No HDFS. Everything is on the normal local disk (`outputs/`).
- VM4 must not copy its file to VM1. It must go through Kafka (topic `order_metrics`).
- Both topics: **2 partitions, replication factor 2**, same cluster.
- Failure test: stop **VM3**, never VM2 (VM2 is the only boss).
- Exact duplicates counted once in both Spark jobs. All calculated numbers rounded to 2 decimals. Nothing hardcoded.

2. Demo day overview

The evaluator checks **4 checkpoints**. Section 3 has only the commands needed for them: first what to run **before** the checkpoints, then each checkpoint's commands in order. Each step says which member runs it; when you finish, say **"done"** and the member with the next step starts.

What each member runs

Member	Before the checkpoints	Checkpoint 1	Checkpoint 2	Checkpoint 3	Checkpoint 4
Member 1 (VM1)	0 setup, 4 clear outputs	-	7 Producer 1	13 Consumer 2, 14 wc -l, 15 Spark Job 2	17 slow producer
Member 2 (VM2)	0 setup, 1 start broker 1, 3 fresh topics	5 cluster checks	6 observer	11 observer	16 describe, 19 describe, 21 describe, 22 election
Member 3 (VM3)	0 setup, 2 start broker 2	-	-	-	18 kill broker, 20 restart broker
Member 4 (VM4)	0 setup, 4 clear outputs	-	8 Consumer 1, 9 wc -l	10 Spark Job 1, 12 Producer 2	-

What the evaluator checks

Checkpoint	Steps	Evidence on screen
1. Brokers reachable, VM2 is the controller, topics exist	5	ids 1 and 2; `LeaderId: 1`; `orders` and `order_metrics`
2. Raw orders flow VM1 -> VM4	6-9	observer prints 3 records; `wc -l` = 62
3. Enriched orders flow VM4 -> VM1	10-15	observer prints 3 records; `wc -l` = 44; the two CSVs (accuracy checked on the portal)
4. Broker failure and recovery	16-22	Isr 1,2 -> kill VM3 -> Leader 1 / Isr 1 while the slow producer keeps sending -> restart -> Isr 1,2 -> election puts the leaders back

Output accuracy is graded from this live run only, on the portal `assignment2eval2026.vercel.app`.

Before you start (everyone)

- `git pull` so every VM has the same code. ZeroTier connected; you can `ping` VM2's ZeroTier IP.
- Use a **Linux / WSL terminal**, not PowerShell. The prompt must look like `user@machine:...$`, not `PS C:\...>`.
- Do step 0 in **every** terminal you'll use.
- Repo scripts are run as `bash kafka/...sh`. If you see `Permission denied` for `./kafka/...`, the copy lost its executable flag: run `chmod +x kafka/*.sh kafka/local/*.sh practice_alone.sh` once, or put `bash` in front.
- Never stop VM2. Don't edit `order.jsonl`.
- Spark's `WARN` lines are normal. Only a `Traceback` means a failure.

3. Demo commands, in order

Demo day: go down the list. Run only the steps tagged with your member number.

Practising alone (one laptop): run **every** step, whatever its tag, using the "Practising alone" box where there is one. Steps 6, 11 and 17 keep running, so do the next step in a second terminal (with step 0 done in it too).

Copying: copy one grey box at a time. A line ending in `\` continues on the next line, so copy both lines together. The broker addresses are already filled in: VM2 = `10.31.88.79:9092`, VM3 = `10.31.88.79:9094`. In the observer commands, replace `teamXX` with your team number.

> **For now VM2's machine also runs VM3's broker** (second broker on port 9094), because Member 3 hasn't joined yet. That machine does the one-time setup box marked "for now" below, then runs both Member 2's and Member 3's steps. Step 18 is `pkill -9 -f broker2.properties` for now, because the TA's `pkill -9 -f kafka.Kafka` would kill both brokers. Member 4 hasn't joined either, so Member 1's laptop runs Member 4's steps too, with Member 1's step 0 box.

One-time setup (the day before, not in front of the TA)

Do this once per machine. It creates the broker config files that steps 1, 2 and 20 use, with the right ZeroTier IPs already in them, so the brokers never announce `localhost`.

Step A · EVERYONE · Make ~/BD-Assignment point at your copy of the repo (Linux VMs)

```
cd ~/distributed-food-ordering/BD-Assignment
ln -sfn "$PWD" ~/BD-Assignment
ls ~/BD-Assignment/kafka
```

Expect: the `ls` lists `local topics.sh vm`. `~/distributed-food-ordering/BD-Assignment` is where the repo is on VM2; on another VM, `cd` to wherever you unzipped or cloned it instead. After this, every `cd ~/BD-Assignment` in this guide works. Member 1 (WSL) skips this.

Step B · MEMBER 2 (VM2) · Broker 1 config, cluster ID, storage, firewall

```
export KAFKA_HOME=~/kafka
cd ~/BD-Assignment
$KAFKA_HOME/bin/kafka-server-stop.sh
while pgrep -f kafka.Kafka > /dev/null; do sleep 1; done
sed -e "s/VM2_ADDRESS/10.31.88.79/g" -e "s#/var/lib/kafka/data#$HOME/kafka-data#" \
    kafka/vm/vm2-broker1.properties > ~/broker1.properties
grep -E "voters|advertised|log.dirs" ~/broker1.properties
$KAFKA_HOME/bin/kafka-storage.sh random-uuid > ~/cluster.id
cat ~/cluster.id
$KAFKA_HOME/bin/kafka-storage.sh format -t "$(cat ~/cluster.id)" -c ~/broker1.properties
sudo ufw allow in on zt+
```

Expect: the `grep` shows `10.31.88.79` (not `VM2_ADDRESS` or `localhost`) and `log.dirs=/home/...`; `cat` prints the cluster ID: **send it to Member 3**; `format` ends with `Formatting ...`. The `ufw` line lets ZeroTier traffic through the firewall. If `format` says the directory is already formatted, run `rm -rf ~/kafka-data` and repeat that line. Box B makes a **new** cluster ID each time, so if you run it again, redo box C too.

Step C · MEMBER 2 (VM2) · For now: broker 2 on the same machine (port 9094)

```
cd ~/BD-Assignment
sed -e "s/VM2_ADDRESS/10.31.88.79/g" -e "s/VM3_ADDRESS:9092/10.31.88.79:9094/" \
    -e "s#0.0.0.0:9092#0.0.0.0:9094#" -e "s#/var/lib/kafka/data#$HOME/kafka-data2#" \
    kafka/vm/vm3-broker2.properties > ~/broker2.properties
grep -E "listeners|voters|log.dirs" ~/broker2.properties
$KAFKA_HOME/bin/kafka-storage.sh format -t "$(cat ~/cluster.id)" -c ~/broker2.properties
```

Expect: the `grep` shows `0.0.0.0:9094`, `10.31.88.79:9094` and `log.dirs=/home/.../kafka-data2`. Then run steps 1 and 2 below on this machine. If `format` says it's already formatted, or box B was run again (new cluster ID), run `rm -rf ~/kafka-data2` and repeat the `format` line. When Member 3 joins, they get their own setup box here and broker 2 moves to their VM.

Before the checkpoints

Step 0 · EVERYONE · Set up the terminal (every terminal you use)

Member 1 (VM1):

```
cd /mnt/c/Users/Aryan/Desktop/BD-Assignment
source .venv-wsl/bin/activate
export KAFKA_HOME=~/kafka
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
unset SPARK_HOME HADOOP_HOME HADOOP_CONF_DIR PYSPARK_PYTHON PYSPARK_DRIVER_PYTHON
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Members 2 and 3 (VM2, VM3):

```
cd ~/BD-Assignment
export KAFKA_HOME=~/kafka
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Member 4 (VM4):

```
cd ~/BD-Assignment
source .venv/bin/activate
export KAFKA_HOME=~/kafka
export B1=10.31.88.79:9092
export B2=10.31.88.79:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Practising alone (one laptop, local cluster):

```
cd /mnt/c/Users/Aryan/Desktop/BD-Assignment
source .venv-wsl/bin/activate
export KAFKA_HOME=~/kafka
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
unset SPARK_HOME HADOOP_HOME HADOOP_CONF_DIR PYSPARK_PYTHON PYSPARK_DRIVER_PYTHON
export B1=localhost:9092
export B2=localhost:9094
export BS=$B1
export BOOTSTRAP=$B1,$B2
export KAFKA_BOOTSTRAP_SERVERS=$B1,$B2
```

Expect: the prompt starts with `(.venv-wsl)` (Member 1) or `(.venv)` (Member 4).

If it fails: `No such file or directory` on the `cd` line: the repo is in another folder, so do one-time box A. `python: command not found` later: this box wasn't run in that terminal, so run it again. A > prompt where nothing happens: part of a line (often a quote) was missed when copying, so press Ctrl+C and copy the whole box again.

Step 1 · MEMBER 2 (VM2) · Start broker 1 and the controller

Demo day:

```
$KAFKA_HOME/bin/kafka-server-start.sh -daemon ~/broker1.properties
```

Practising alone (one laptop, local cluster):

```
bash kafka/local/cluster.sh format
bash kafka/local/cluster.sh start 1
```

Expect: no output. (Alone: `format` is needed after every WSL restart and is safe to repeat.)

If it fails: if the broker doesn't come up and `$KAFKA_HOME/logs/kafkaServer.out` says `class file versions up to 55.0`, Java is too old: run `export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64` and start it again.

Step 2 · MEMBER 3 (VM3) · Start broker 2

Demo day:

```
$KAFKA_HOME/bin/kafka-server-start.sh -daemon ~/broker2.properties
```

Practising alone (one laptop, local cluster):

```
bash kafka/local/cluster.sh start 2
```

Expect: no output.

Step 3 · MEMBER 2 (VM2) · Create fresh topics

```
bash kafka/topics.sh delete
sleep 5
bash kafka/topics.sh create
```

Expect: both topics with `PartitionCount: 2`, `ReplicationFactor: 2`, Isr 1,2.

If it fails: `Permission denied`: type it with `bash` in front, as here (or run `chmod +x kafka/*.sh kafka/local/*.sh` once). `Connection to node ... could not be established`: a broker isn't running, so redo steps 1 and 2.

Step 4 · MEMBERS 1 + 4 (VM1 / VM4) · Clear old outputs

```
rm -rf outputs/*.jsonl outputs/*.csv outputs/failover_test outputs/recovery_test
mkdir -p outputs/failover_test outputs/recovery_test
```

Expect: no output. The TA only accepts outputs made during the live run.

Checkpoint 1: cluster

Step 5 · MEMBER 2 (VM2) · Brokers reachable, VM2 is the controller, topics exist

```
$KAFKA_HOME/bin/kafka-broker-api-versions.sh --bootstrap-server "$BS" | grep "id:"
$KAFKA_HOME/bin/kafka-metadata-quorum.sh --bootstrap-server "$BS" describe --status
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server "$BS" --list
```

Expect: two brokers, `id: 1` and `id: 2`; `LeaderId: 1` (VM2 is the controller); `orders` and `order_metrics`.

If it fails: the brokers are shown as `localhost`: the configs from one-time boxes B and C aren't being used, so redo them, then steps 1 and 2. `Connection to node -1 ... could not be established` or `Connection refused`: broker 1 isn't running, so run step 1, wait 10 s, then step 2. `timed out`: the broker machine's firewall is blocking, so run `sudo ufw allow in on zt+` there.

Checkpoint 2: raw orders, VM1 -> Kafka -> VM4

Step 6 · MEMBER 2 (VM2) · Observer on orders (leave it running)

```
$KAFKA_HOME/bin/kafka-console-consumer.sh --bootstrap-server "$BS" \
    --topic orders --group ta-observer-teamXX-run1 \
    --consumer-property auto.offset.reset=latest \
    --max-messages 3 --timeout-ms 30000
```

Expect: it waits, and gives up after 30 s, so step 7 must start right away. The `deprecated` warning is harmless.

If it fails: it ends with `Processed a total of 0 messages`: step 7 started too late. Run this again, then step 7 straight away.

Step 7 · MEMBER 1 (VM1) · Producer 1

```
python producer/producer1.py --input order.jsonl
```

Expect: `published 62 records`; the observer prints 3 records and `Processed a total of 3 messages`.

If it fails: `python: command not found`: run your step 0 box in this terminal. `NoBrokersAvailable` or `KafkaTimeoutError`: this laptop can't use the brokers; step 5 must show `10.31.88.79` for both brokers (not `localhost`), and step 0 must have been run here.

Step 8 · MEMBER 4 (VM4) · Consumer 1

```
python consumer/consumer1.py --output outputs/orders_landed.jsonl
```

Expect: `landed 62 records`.

Step 9 · MEMBER 4 (VM4) · Show the landed file

```
wc -l outputs/orders_landed.jsonl && head -n 2 outputs/orders_landed.jsonl
```

Expect: `62 outputs/orders_landed.jsonl` and two order lines.

Checkpoint 3: enriched orders, VM4 -> Kafka -> VM1

Step 10 · MEMBER 4 (VM4) · Spark Job 1

```
python spark/spark_job1.py --input outputs/orders_landed.jsonl \
    --output outputs/order_metrics.jsonl
```

Expect: `enriched records written: 44`.

If it fails: `'JavaPackage' object is not callable`: an old Spark install is being picked up; run the `unset ...` line from Member 1's step 0 box, then run this again.

Step 11 · MEMBER 2 (VM2) · Observer on order_metrics (leave it running)

```
$KAFKA_HOME/bin/kafka-console-consumer.sh --bootstrap-server "$BS" \
    --topic order_metrics --group ta-observer-teamXX-run2 \
    --consumer-property auto.offset.reset=latest \
    --max-messages 3 --timeout-ms 30000
```

Expect: it waits; step 12 must start within 30 s.

If it fails: it ends with `Processed a total of 0 messages`: step 12 started too late. Run this again, then step 12 straight away.

Step 12 · MEMBER 4 (VM4) · Producer 2

```
python producer/producer2.py --input outputs/order_metrics.jsonl
```

Expect: `published 44 records`; the observer prints 3 records.

Step 13 · MEMBER 1 (VM1) · Consumer 2

```
python consumer/consumer2.py --output outputs/order_metrics_landed.jsonl
```

Expect: `landed 44 records`.

Step 14 · MEMBER 1 (VM1) · Show the landed file

```
wc -l outputs/order_metrics_landed.jsonl && head -n 2 outputs/order_metrics_landed.jsonl
```

Expect: `44 outputs/order_metrics_landed.jsonl` and two enriched lines.

Step 15 · MEMBER 1 (VM1) · Spark Job 2 (the final CSVs)

```
python spark/spark_job2.py --input outputs/order_metrics_landed.jsonl \
    --output-dir outputs
```

Expect: `6 rows -> restaurant_summary.csv` and `4 rows -> city_summary.csv`. If the TA wants to see them: `cat outputs/restaurant_summary.csv outputs/city_summary.csv`.

If it fails: `'JavaPackage' object is not callable`: an old Spark install is being picked up; run the `unset ...` line from Member 1's step 0 box, then run this again.

Checkpoint 4: broker failure and recovery

Step 16 · MEMBER 2 (VM2) · Leaders and ISR before the failure

```
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server "$BS" --describe --topic orders
```

Expect: `Isr: 1,2` / `2,1` on both partitions.

Step 17 · MEMBER 1 (VM1) · Slow producer (leave it running)

```
python producer/producer1.py --file order.jsonl --delay 0.5
```

Expect: one line per record, `sent 1/62 -> partition ...`, every half second. Keep it on screen: after the kill in step 18 it pauses briefly, then carries on to `sent 62/62` and `published 62 records` with no error. That is the pipeline continuity the TA checks.

Step 18 · MEMBER 3 (VM3) · Kill broker 2 (while step 17 is still sending)

```
pkill -9 -f broker2.properties
```

Expect: no output; only broker 2 dies. **Not** the TA's `pkill -9 -f kafka.Kafka` here: on this shared machine it would kill broker 1 and the controller as well. Once VM3 is its own machine, the TA's command is the one to use there.

Step 19 · MEMBER 2 (VM2) · Leaders and ISR after the failure

```
sleep 10
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server "$BS" --describe --topic orders
```

Expect: `Leader: 1` and `Isr: 1` on both partitions. A `WARN ... Connection to node 2 ... could not be established` line is normal here: broker 2 is the one that's down.

If it fails: the describe never finishes and every node fails: both brokers died (the TA's `pkill -9 -f kafka.Kafka` on a machine that runs both). Run steps 1 and 2, then redo from step 16.

Step 20 · MEMBER 3 (VM3) · Restart the broker in a second terminal (leave it open)

Demo day:

```
KAFKA_HEAP_OPTS='-Xmx512m -Xms512m' $KAFKA_HOME/bin/kafka-server-start.sh \
    ~/broker2.properties
```

Practising alone (one laptop, local cluster):

```
bash kafka/local/cluster.sh start 2
```

Expect: it keeps printing log lines (some `WARN` ones are normal), including `Kafka Server started`, and doesn't return to the prompt. Run it in a **separate terminal** (with step 0 done) and don't close it or press Ctrl+C: that stops broker 2 again.

If it fails: it returns to the prompt by itself: broker 2 didn't start; the last lines it printed say why.

Step 21 · MEMBER 2 (VM2) · Leaders and ISR after the restart

```
sleep 15
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server "$BS" --describe --topic orders
```

Expect: `Isr: 1,2` again; leaders are still all 1. Don't go on to step 22 until it says so.

If it fails: still `Isr: 1`: broker 2 isn't running. Check the step 20 terminal is still open and printing; if not, run step 20 again, wait 15 s, and repeat this step.

Step 22 · MEMBER 2 (VM2) · Leader election, then check again

```
$KAFKA_HOME/bin/kafka-leader-election.sh --bootstrap-server "$BS" \
    --election-type preferred --all-topic-partitions
$KAFKA_HOME/bin/kafka-topics.sh --bootstrap-server "$BS" --describe --topic orders
```

Expect: `Successfully completed leader election`; the leaders are back to how they were in step 16. **Demo finished.** (Alone: `bash kafka/local/cluster.sh stop all`.)

If it fails: many `PreferredLeaderNotAvailableException` lines and `Isr: 1` in the describe: broker 2 isn't back yet (the preferred leader of partition 0 is broker 2). Do step 20 again in its own terminal, wait until step 21 shows `Isr: 1,2`, then run this again. `Connection to node ... could not be established` and it never finishes: broker 1 is down too, so run step 1, wait 10 s, run step 2, then steps 21 and 22 again.

4. Member 1 (VM1): Producer 1, Consumer 2, Spark Job 2

Member 1 runs three programs: **producer1.py** at the start, then later **consumer2.py** and **spark_job2.py** at the end. Two shared helper files sit behind them: `publish.py` and `landing.py`.

4.1 producer1.py: post the raw orders

Reads `order.jsonl` line by line and sends each line to topic **orders**. All of the real work is in the shared helper `publish.py`.

- **Sends every line as-is, junk included.** The producer only delivers; cleaning is Spark Job 1's job. If both did some cleaning, the rules would be split across two places.
- **No key.** To get a key (order_id) it would have to read the JSON, and some lines are broken. Without a key Kafka spreads messages randomly over the 2 partitions (e.g. 31 / 31, and the split changes each run).
- **acks="all"**: Kafka only says "received" once *every* in-sync copy has the message. This is the safest setting.
- **retries=10**: if something hiccups (e.g. VM3 goes down mid-send), it tries again automatically.
- **linger_ms=10**: waits up to 10 ms to bundle messages together, so fewer network trips.
- It waits for confirmation of every message (`flush()` + `f.get()`), so the printed "62 sent" means 62 *confirmed*, along with which partition each one landed in.

4.2 consumer2.py + landing.py: collect the cleaned orders

Reads *everything* from topic **order_metrics** and saves it to `order_metrics_landed.jsonl` (44 lines). Member 4's consumer1.py is the same code with topic **orders**.

- **Reads all partitions itself** (`assign()`, no consumer group). A consumer group remembers where it stopped, so a second run would get nothing. We want the *whole* topic every time.
- **Always starts from the beginning** (`seek_to_beginning`, offset 0).
- **Knows when to stop.** Normal consumers run forever. Ours stops once it has reached the last message of every partition *and* nothing new has arrived for 10 s (`--idle-timeout`).
- **Can start early.** `--wait-timeout` (default 120 s) lets it wait for the producer.
- **Overwrites the file** (`"w"` mode), so running it twice doesn't double the data.
- Prints how many records came from each partition: **16 from partition 0, 28 from partition 1**.

4.3 spark_job2.py: make the two reports

Reads `order_metrics_landed.jsonl` and writes **restaurant_summary.csv** (6 rows) and **city_summary.csv** (4 rows).

Step	What happens	Why
1. Read	Read the file with a fixed schema (types given up front)	Types are guaranteed and the file is read only once
2. Dedup	`dropDuplicates` on all 15 columns	Spec: exact duplicates count once. Removes 0 here because Job 1 already did it, but it protects against a producer being run twice
3. Group	groupBy restaurant_id, then groupBy city	The two analytics the spec asks for
4. Compute	count, averages, delayed count (rows where status = DELAYED), delay_rate_pct = 100 x delayed / count	delay_rate is worked out from the exact counts, and rounded only once at the end
5. Round	`F.round(..., 2)` on every calculated number	Spec rule

Step	What happens	Why
6. Sort + write	Sort the rows, collect them (just 10 rows) and write with Python's csv module	Spark's own writer makes a *folder* of part-files, but the spec wants one file with an exact name

`.cache()` keeps the deduplicated data in memory because it is used by two reports. Without it Spark would recompute it for each one.

4.4 Final outputs (know these by heart)

restaurant	orders	avg amount	avg prep	avg delivery	delayed	delay %
R01	8	548.5	13.13	22.0	4	50.0
R02	8	539.25	13.38	27.0	8	100.0
R03	7	664.0	13.86	32.0	4	57.14
R04	7	639.57	12.14	37.0	7	100.0
R05	7	615.14	12.0	42.0	3	42.86
R06	7	712.14	13.43	47.0	7	100.0

city	orders	avg amount	avg delivery	delayed	delay %
Bengaluru (R01+R02)	16	543.88	24.5	12	75.0
Hyderabad (R06)	7	712.14	47.0	7	100.0
Mumbai (R03+R04)	14	651.79	34.5	11	78.57
Pune (R05)	7	615.14	42.0	3	42.86

Easy cross-check: each restaurant is in exactly one city, so Pune = R05 and Hyderabad = R06. Bengaluru = R01 + R02: 8+8 = 16 orders, 4+8 = 12 delayed, 12/16 = 75%. Order counts add up to 44; delayed orders add up to 33 of 44. Example: R05 delay % = 100 x 3 / 7 = 42.857 -> **42.86**.

Your steps (section 3): 0, 4 before; 7; 13, 14, 15; 17.
Show the evaluator: 62 published (TA observer sees 3); 44 landed and `wc -l` = 44; 6 + 4 rows, verify OK; the slow producer reaching 62/62 while VM3 is killed.
Be ready to explain: why Producer 1 has no key; why the consumer reads every partition from offset 0 and stops by itself; how Job 2 dedupes, groups and rounds; why the producer survives the failure (acks=all, retries, min.insync.replicas=1).

5. Member 2 (VM2): Broker 1 + KRaft controller (the boss)

- Runs Kafka with `process.roles=broker,controller`, `node.id=1`.
- **KRaft** = Kafka manages itself, with no separate ZooKeeper. The controller keeps the cluster's "register" (which brokers are alive, which topics exist, who leads each partition).
- Port **9092** = normal Kafka traffic, port **9093** = controller traffic (only VM2 has it).
- Generates the **cluster ID** once; VM3 must use the same ID to join.
- `min.insync.replicas=1`: writes still succeed with only one copy alive (needed while VM3 is off).
- `auto.create.topics.enable=false`: topics are only created on purpose, so a typo can't create a wrong topic.
- In the demo, Member 2 shows the cluster state: both brokers, the topic config, and the ISR before, during and after the failure (section 8 explains how to read it).

Your steps (section 3): 0, 1, 3 before; 5; 6; 11; 16, 19, 21, 22. You run the TA's cluster commands and both observers.
Show the evaluator: both broker ids; `LeaderId: 1` (VM2 is the controller); the two topics; the observers printing 3 records each; Isr 1,2 -> Leader 1 / Isr 1 -> Isr 1,2 -> leaders restored by the election.
Be ready to explain: KRaft and what the controller does; node.id, listeners vs advertised.listeners, ports 9092/9093; cluster ID; min.insync.replicas=1; Leader vs Replicas vs Isr; preferred leader election; why VM2 can't be stopped.

6. Member 3 (VM3): Broker 2 (the backup)

- Runs Kafka with `process.roles=broker` only, `node.id=2`.
- Points at VM2:9093 to join and is formatted with **VM2's cluster ID** (it must not create its own cluster).
- Holds the second copy of every partition. In the failure test, **this is the machine that gets stopped**.
- When it stops, VM2's controller notices the missing heartbeats, removes it from the ISR and makes broker 1 the leader everywhere. When it restarts, it copies what it missed and rejoins the ISR.

Your steps (section 3): 0, 2 before; 18, 20. Your broker is the one that fails.
Show the evaluator: the broker being killed with `pkill -9` while the slow producer runs, then restarted in the foreground. Then point to Member 2's describe output, where Isr shrinks to 1 and grows back to 1,2.
Be ready to explain: what happens step by step when VM3 stops (heartbeats stop, it's fenced, leadership moves to broker 1); how it catches up and rejoins the ISR; why VM3 uses VM2's cluster ID; what RF 2 protects against.

7. Member 4 (VM4): Consumer 1, Spark Job 1, Producer 2 (the cleaning step)

- **consumer1.py**: same code as consumer2, but reads topic **orders** -> `orders_landed.jsonl` (62 lines).
- **spark_job1.py**: the heart of the project. It **throws away bad records**, **removes duplicates**, and **adds 7 new columns**. Output: `order_metrics.jsonl` (44 lines).
- **producer2.py**: posts those 44 lines to topic **order_metrics**, **keyed by order_id**. The same key always goes to the same partition, so the split is always **16 / 28**.

Cleaning funnel: 62 -> 48 -> 44

Thrown out because...	Count
status is not DELIVERED (T04X0001-6: cancelled, pending, failed, refunded)	6
no order_id (2 "invalid_order_record" lines)	2
missing fields ({"order_id":"BROKEN"} and a T040002 copy without a city)	2
bad timestamp (T040005 "bad", T040003 "not-a-timestamp")	2
distance_km <= 0 (T040001: -2) and order_amount <= 0 (T040004: 0)	2
Total dropped -> 48 valid	**14**
Exact duplicates removed (T040006, T040017, T040028, T040039) -> **44**	4

New columns Spark Job 1 adds (all in minutes, rounded to 2 decimals):

Column	Formula
preparation_time_min	picked_up - accepted
delivery_time_min	delivered - picked_up
fulfillment_time_min	delivered - order_timestamp
delivery_speed_kmph	distance_km / (delivery_time_min / 60)
delay_min	max(0, delivered - expected)
delivery_status	ON_TIME if delay = 0, else DELAYED
delay_category	0 ON_TIME | up to 15 SLIGHT | up to 30 MODERATE | over 30 SEVERE

Midnight trick: it uses full date+time, not just the time of day. T040023 picked up 23:20 and delivered 00:02 the next day = **42 min**, not -1398. The time zone is set to UTC so daylight saving can't shift anything.

Your steps (section 3): 0, 4 before; 8, 9; 10, 12.
Show the evaluator: 62 landed and `wc -l` = 62; the drop report (14 dropped, 48 valid, 4 duplicates, 44 written); 44 published, split 16 / 28 (TA observer sees 3).
Be ready to explain: each cleaning rule and the 62 -> 48 -> 44 funnel; the 7 new columns and the midnight case (T040023 = 42 min); why Producer 2 is keyed by order_id.

8. The failure test explained (whole team)

```
Before (both up):    orders P0  Leader: 2  Replicas: 2,1  Isr: 2,1
                     orders P1  Leader: 1  Replicas: 1,2  Isr: 1,2
After stopping VM3:  orders P0  Leader: 1  Replicas: 2,1  Isr: 1     <- leader moved to 1
                     orders P1  Leader: 1  Replicas: 1,2  Isr: 1
Restart VM3 -> Isr back to 2,1 / 1,2.
```

Leader = the broker currently serving that partition. **Replicas** = where copies are *supposed* to live (doesn't change). **Isr** = copies that are currently up to date (shrinks to just 1 while VM3 is off).

Producing still works with VM3 down because the producer uses `acks=all` (wait for every *in-sync* copy, now just broker 1) and the brokers allow `min.insync.replicas=1`. While the leader moves (about 10 s after the kill) sends to that partition fail briefly; the producer retries every second for up to 30 tries, so no record is lost.

Preferred leader election (step 22): each partition's *preferred* leader is the first broker in its `Replicas` list. After VM3 comes back, broker 1 is still leading everything, so the load is uneven. `kafka-leader-election.sh --election-type preferred` moves leadership back to the preferred brokers straight away (Kafka would also do it by itself after a few minutes).

kill -9 vs a normal stop: `pkill -9` kills the broker instantly, like a crash or power cut, so it can't hand over its partitions politely. The controller only notices when the broker's heartbeats stop.

9. Key words in one line each

Term	Simple meaning
Broker	A Kafka server that stores messages (VM2, VM3).
Controller	The boss broker that tracks who is alive and picks leaders (only VM2).
KRaft	Kafka's built-in way of electing and storing the boss's records. It replaced ZooKeeper.
Topic	A named mailbox: *orders*, *order_metrics*.
Partition	A topic is split into pieces (we use 2) so work can be shared. Order is kept only inside one partition.
Offset	A message's position number inside a partition (0, 1, 2 ...).
Replication factor 2	Every partition is stored twice, once on each broker.
Leader / follower	The leader copy handles reads and writes; the follower copies it.
ISR	In-Sync Replicas: the copies that are fully up to date.
acks=all	The producer gets "OK" only when all in-sync copies have the message.
Bootstrap servers	The first broker address(es) a client contacts to discover the whole cluster.
advertised.listeners	The address a broker tells clients to use. It must be the ZeroTier IP, not localhost.
ZeroTier	A virtual LAN, so the 4 laptops on different home networks can reach each other.
local[*]	Spark runs on this one machine, using all CPU cores.
Lazy evaluation	Spark only plans the work; it runs when an action (count, collect) is called.

10. Top viva questions, short answers

Q. Why use Kafka instead of just copying files?

It separates the stages (VM1 and VM4 never talk directly), stores the data reliably in two copies, and lets any stage re-read the data. The spec also forbids copying the file.

Q. Why is Producer 1 unkeyed but Producer 2 keyed?

Raw lines may be broken JSON, so Producer 1 doesn't read them to find a key. Cleaned lines are always valid, so Producer 2 keys by order_id, which gives a fixed partition split (16/28).

Q. Why doesn't the consumer use a consumer group?

Groups remember where they stopped, so a second run would read nothing. Ours is a batch job that must copy the whole topic every time, so it reads all partitions from offset 0 and commits nothing.

Q. How does the consumer know when to stop?

When its position equals the last offset of every partition and nothing new has arrived for 10 s.

Q. What happens when VM3 stops?

VM3 stops sending heartbeats, the controller on VM2 removes it from the ISR and makes broker 1 the leader everywhere. Broker 1 already has a full copy (RF 2), so no data is lost.

Q. Why stop VM3 and not VM2?

VM2 is the only controller. Without it nobody can elect leaders. Real high availability needs 3 controllers.

Q. Can you still produce while VM3 is down?

Yes. acks=all only waits for the in-sync copies (now just broker 1), and min.insync.replicas=1 allows that.

Q. What if Producer 1 runs twice?

The topic has 124 messages, but Spark Job 1 still outputs 44 because duplicates are exact and get removed. Delivery is at-least-once, which is why both jobs dedupe.

Q. Why collect() and write the CSV with Python?

Spark's writer makes a folder of part-files. The spec wants single files with exact names, and the results are tiny (6 and 4 rows).

Q. Why is dropDuplicates in Job 2 if it removes 0?

The spec says both jobs must dedupe. It also protects against Kafka redelivery or a producer re-run.

Q. Why not deduplicate by order_id?

The spec says *exact* duplicates. Choosing between two different records with the same ID would be inventing a business rule.

Q. How do you know the outputs are correct?

tests/verify_outputs.py redoes everything in plain Python (no Spark) and compares every value. It prints "OK: 44 enriched records and both summaries match".

Q. Why Spark for only 62 records?

The assignment is about the architecture. The same code would work on millions of rows or on a real cluster.

11. Numbers to remember

Item	Value
Raw lines	62
Dropped by cleaning	14 -> 48 valid
Exact duplicates removed	4 -> 44 enriched
orders partition split	random, e.g. 31 / 31
order_metrics partition split	always 16 / 28
Restaurants / cities	6 (R01-R06) / 4 (Bengaluru, Hyderabad, Mumbai, Pune)

Item	Value
Delayed orders	33 of 44 (11 on time)
Delay categories	ON_TIME 11, SLIGHT 11, MODERATE 15, SEVERE 7
Highest / lowest delay %	100% (R02, R04, R06, Hyderabad) / 42.86% (R05, Pune)
Fastest / slowest delivery	R01 22.0 min / R06 47.0 min
Node IDs / ports	VM2 = node 1 (9092 + 9093); VM3 = node 2 (9092)
Versions	Kafka 4.3.1 (KRaft), PySpark 4.2.0, kafka-python 2.2.15, Java 17+